

THE TWENTY-FIRST WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

December 3, 1960

Morning Session

1. Let n be a given positive integer. How many solutions are there in ordered positive integer pairs (x, y) to the equation

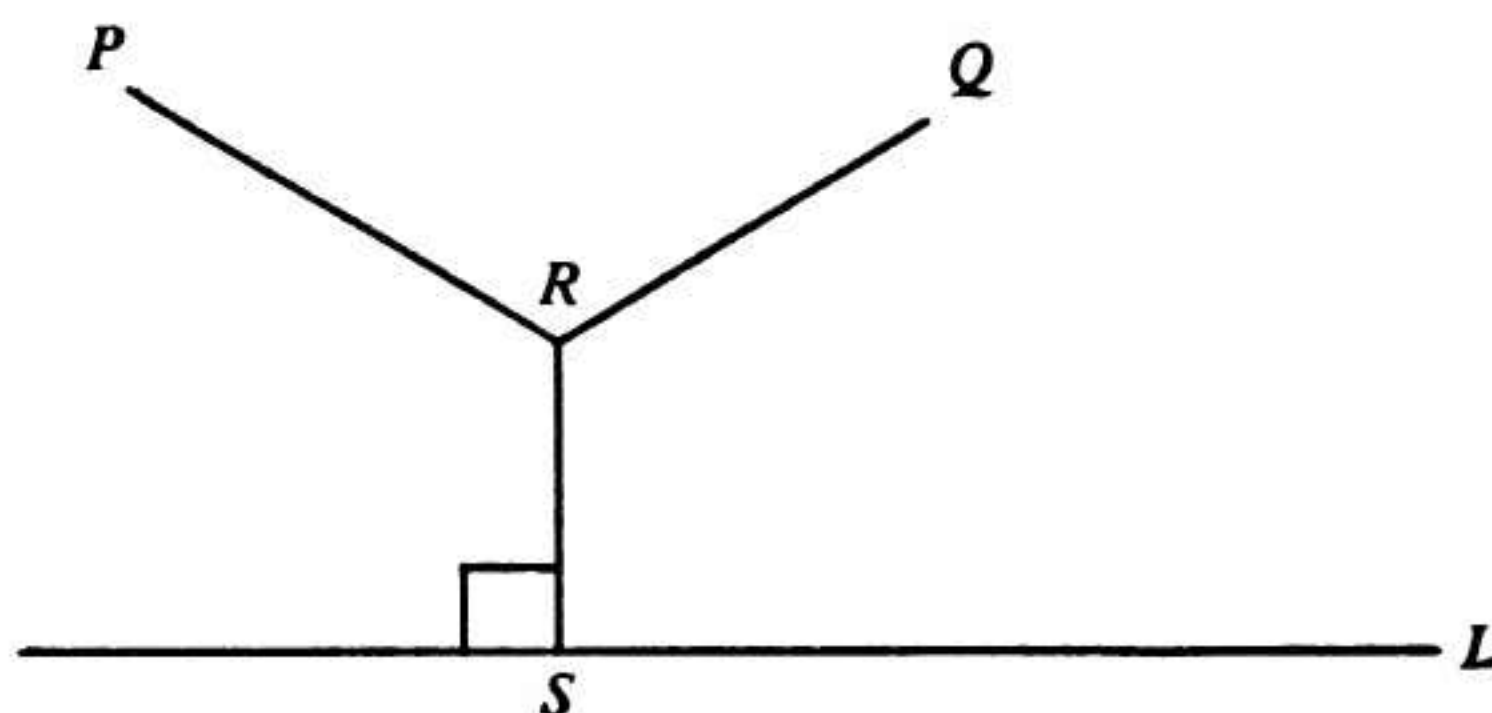
$$\frac{xy}{x+y} = n? \quad (\text{page 516})$$

2. Show that if three points are inside a closed square of unit side, then some pair of them are within $\sqrt{6} - \sqrt{2}$ units apart. (page 516)

3. Show that if t_1, t_2, t_3, t_4, t_5 are real numbers, then

$$\sum_{j=1}^5 (1 - t_j) \exp \left(\sum_{k=1}^j t_k \right) \leq e^{e^{e^e}}. \quad (\text{page 518})$$

4. Given two points in the plane, P and Q , at fixed distances from a line L , and on the same side of the line, as indicated, the problem is to find a third point R so that $PR + RQ + RS$ is a minimum, where RS is perpendicular to L . Consider all cases.



(page 519)

5. Consider a polynomial $f(x)$ with real coefficients having the property $f(g(x)) = g(f(x))$ for every polynomial $g(x)$ with real coefficients. Determine and prove the nature of $f(x)$. (page 521)

6. A player throwing a die scores as many points as on the top face of the die and is to play until his score reaches or passes a total n . Denote by $p(n)$ the probability of making exactly the total n , and find the value of $\lim_{n \rightarrow \infty} p(n)$. (page 522)

7. Let $N(n)$ denote the smallest positive integer N such that $x^N = 1$ for every permutation x on n symbols, where 1 denotes the identity permutation. Prove that if $n > 1$,

$$\begin{aligned} \frac{N(n)}{N(n-1)} &= 1 \text{ if } n \text{ is divisible by 2 distinct primes,} \\ &= p \text{ if } n \text{ is a power of a prime } p. \end{aligned} \quad (\text{page 526})$$

Afternoon Session

1. Find all solutions of $n^m = m^n$ in integers n and m ($x \neq m$). Prove that you have obtained all of them. (page 527)

2. Evaluate the double series

$$\sum_{j=0}^{\infty} \sum_{k=0}^{\infty} 2^{-3k-j-(k+j)^2}. \quad (\text{page 528})$$

3. The motion of the particles of a fluid in the plane is specified by the following components of velocity

$$\begin{aligned} (1) \quad \frac{dx}{dt} &= y + 2x(1 - x^2 - y^2), \\ \frac{dy}{dt} &= -x. \end{aligned}$$

Sketch the shape of the trajectories near the origin. Discuss what happens to an individual particle as $t \rightarrow +\infty$, and justify your conclusion. (page 529)

Note. The second y in the first equation above was broken and looked like a v on the examination as it was distributed to the contestants.

4. Consider the arithmetic progression $a, a + d, a + 2d, \dots$, where a and d are positive integers. For any positive integer k , prove that the progression has either no exact k th powers or infinitely many. (page 532)

5. Define a sequence as follows:

$$a_0 = 0$$

$$a_1 = 1 + \sin(-1)$$

$$\vdots$$

$$a_n = 1 + \sin(a_{n-1} - 1)$$

$$\vdots$$

Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n a_k. \quad (\text{page 534})$$

6. Any positive integer may be written in the form $n = 2^k(2l + 1)$. Let $a_n = e^{-k}$ and $b_n = a_1 a_2 a_3 \cdots a_n$. Prove that $\sum b_n$ converges. (page 535)

7. Let $g(t)$ and $h(t)$ be real, continuous functions for $t \geq 0$. Show that any function $v(t)$ satisfying the differential inequality

$$\frac{dv}{dt} + g(t)v \geq h(t), \quad v(0) = c,$$

satisfies the further inequality $v(t) \geq u(t)$ where

$$\frac{du}{dt} + g(t)u = h(t), \quad u(0) = c.$$

From this, conclude that for sufficiently small $t > 0$, the solution of

$$\frac{dv}{dt} + g(t)v = v^2, \quad v(0) = c_1$$

may be written

$$v = \max_w \left[c_1 e^{-\int_0^t [g(s) - 2w(s)] ds} - \int_0^t e^{-\int_0^s [g(s_1) - 2w(s_1)] ds_1} w^2(s) ds \right]$$

where the maximization is over all continuous functions $w(t)$ defined over some t -interval $[0, t_0]$. (page 536)